Observability

When systems are distributed, **failures are inevitable**. The key is being able to detect and resolve issues **quickly**, before they impact overall performance or damage your organization's reputation.

Observability's ultimate goal is visibility into your system:

- It allows you to understand what's happening **inside your software from the outside**.
- It equips you with the data needed to answer everyday operational questions and to **trace failures to their root causes**.

An effective observability strategy answers questions like:

- Why is service "y" failing?
- What went wrong during the deployment of feature "x"?
- Why has system performance degraded recently?
- What was the state of my service at time "y"?
- Is the issue affecting all users or only a subset?

Observability in DevOps

Observability is essential for DevOps and SRE, enabling teams to **see inside complex systems** and act on insights. Key benefits include:

- **Continuous Deployment** – detect issues immediately after release.
- **Capacity Planning** – identify bottlenecks and scale resources proactively.
- **Incident Response & Retrospectives** – analyze root causes and prevent recurrence.
- **Cultural & Process Benefits** – supports collaboration and a data-driven, blameless culture.

SLI-SLO & Observability

Observability is not just about **what happens**, but also about **how well the system meets expectations**.

- **SLI (Service-Level Indicator)** – A quantitative measure of system health.
  - Percentage of HTTP requests with latency < 500ms.
  - Error rate for payment transactions.
- **SLO (Service-Level Objective)** – The target goal for an SLI.
  - 99.9% of requests must meet the latency target.
  - Payment transaction success rate must remain above 99.5%.

Security & Observability

Observability is not just about performance and reliability — it is also a **powerful security tool**. Integrating security considerations into your observability strategy helps detect threats, ensure compliance, and respond to incidents faster.

Audit Trails

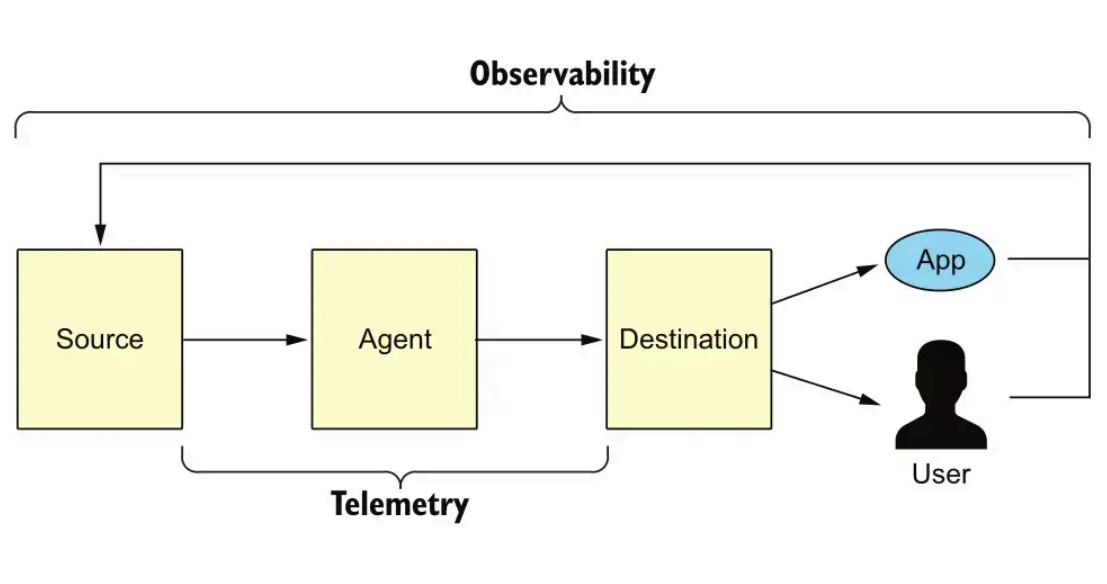
- Track all critical operations and access events.
- **Benefits:**
  - Regulatory compliance (GDPR, PCI-DSS, HIPAA).
  - Forensic investigation in case of incidents.
- Can include centralized logs, traces, and system message metadata.

Correlation Across Signals

- Sophisticated attacks generate multiple signals across the system simultaneously:
  - **Metrics:** unusual CPU, memory, or network usage.
  - **Logs:** failed authentication, suspicious access attempts.
  - **Traces:** abnormal service-to-service calls.
- Correlating metrics, logs, and traces helps to **reconstruct attacks** and identify the entry point.

Pillars of Observability

- **Sources**: Part of the infrastructure and application layer, such as a microservice, a device, a database, a message queue, or the operating system. They typically must be instrumented to emit signals.
- **Signals**: Information emitted by sources. There are different signal types, the most common are:
  - logs
  - metrics
  - traces
- **Agents**: Responsible for signal collection, processing, and routing.
- **Destinations**: Where you consume signals, for different reasons and use cases. These include:
  - visualizations (e.g., dashboards)
  - alerting
  - long-term storage (for regulatory purposes)
  - analytics (finding new usages for an app)
- **Telemetry**: The process of collecting signals from sources, routing or preprocessing via agents, and ingestion to destinations.


Observability
Source
Agent
Destination
App
User
Telemetry

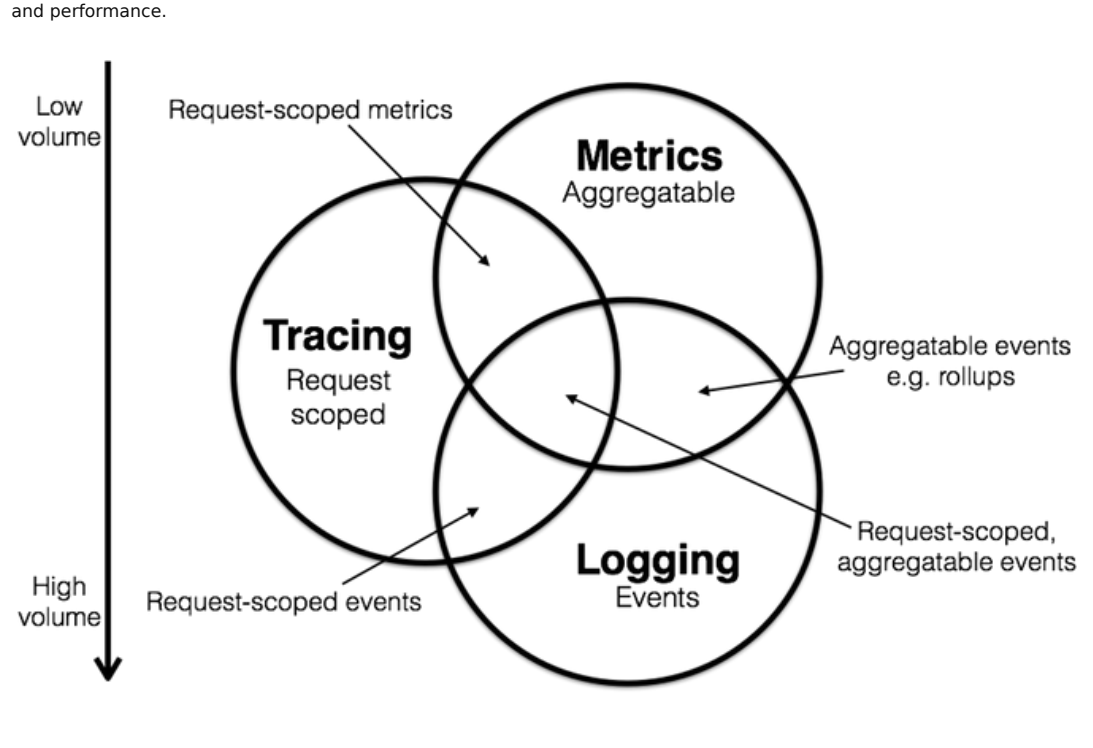
Sources

We categorize sources using a simple and widely used trichotomy: **compute**, **storage**, and **network**. This broad categorization helps understand where issues arise, how they propagate through the system, and what steps to take to maintain health of distributed architectures.

Category	Description	Specific Metrics
Compute	The environments where code runs, including virtual machines, containers, and serverless functions.	- **Container-level metrics**: CPU usage, memory utilization, and resource limits from Docker/Kubernetes. - **Function metrics**: Execution time, cold start duration, and memory consumption for serverless functions (e.g., AWS Lambda, Google Cloud Functions).
Storage	Systems that persist and manage data, such as databases, file systems, and object storage.	- **Database metrics**: Queries per second (QPS), read/write latency, connection pool utilization, rollback/error rates. - **Object storage metrics**: S3 `GetObject` and `PutObject` latency, success/failure rates, storage throughput.
Network	Components that enable communication between services, monitoring connectivity, latency, and throughput.	- **API Gateway metrics**: Request/response latency, success/failure rates, throttled requests. - **Load Balancer metrics**: Request count, 5xx error rates (`HTTPCode_ELB_5XX_Count`), latency. - **Network traffic metrics**: Bandwidth usage, packet loss, data transfer rates.

Signals

The three pillars of observability—**metrics**, **logs**, and **traces**—play a vital role in providing insights into the system's behavior and performance.


Low volume
High volume
Request-scoped metrics
Metrics
Aggregatable
Tracing
Request scoped
Aggregatable events e.g. rollups
Request-scoped, aggregatable events
Logging
Events
Request-scoped events

Metrics

Metrics are numerical values that capture key performance indicators (KPIs) about your system over time. They are typically aggregated and provide an overview of system health and performance.

- **Quantitative Data**: Metrics are quantitative and can be counted or measured (e.g., CPU, memory, disk, threads, latency, etc.).
- **Time-Series Nature**: Metrics are usually collected as time-series data and plotted on dashboards to show trends over time.

Types:

- **Counter** – A monotonically increasing value that resets only on restart. It is used for counting occurrences, such as the number of HTTP requests or errors.
- **Gauge** – A value that can increase or decrease, used for measuring things like memory usage or temperature.
- **Histogram** – A metric that samples observations and counts them in configurable buckets, used for tracking request durations or response sizes. It also provides a total count and sum of values.

Labels: Metrics often include **labels** (also called tags or dimensions) to add additional context. Labels are key-value pairs that allow you to break down metrics by categories such as service, endpoint, status code, region, or instance. For example:

```
http_requests_total{method="GET", endpoint="/api/users", status="200", region="eu-west-1"}
```

Here, `http_requests_total` is the metric name, and the labels provide context about the request method, endpoint, HTTP status, and geographic region. Labels make it easier to filter, aggregate, and slice metrics in dashboards and alerts.

```
http://localhost:8080/actuator/prometheus

# HELP http_server_requests_seconds
# TYPE http_server_requests_seconds summary
http_server_requests_seconds_count{error="none",exception="none",method="GET",outcome="SUCCESS",status="200",u
1
http_server_requests_seconds_sum{error="none",exception="none",method="GET",outcome="SUCCESS",status="200",uri
0.01236351
http_server_requests_seconds_count{error="none",exception="none",method="GET",outcome="SUCCESS",status="200",u
4
http_server_requests_seconds_sum{error="none",exception="none",method="GET",outcome="SUCCESS",status="200",uri
0.016967813
http_server_requests_seconds_count{error="none",exception="none",method="GET",outcome="SUCCESS",status="200",u
10
http_server_requests_seconds_sum{error="none",exception="none",method="GET",outcome="SUCCESS",status="200",uri
0.14632263
```

> ***`http_server_requests_seconds_count`***
>
> - ***Counts** the number of HTTP requests served with these characteristics (GET, endpoint `/actuator`, status 200, no error).*
> - ***Value**: `1` → 1 request served.*
>
> ***`http_server_requests_seconds_sum`***
>
> - ***Total time** spent serving the same requests, in seconds.*
> - ***Value**: `0.01236351` → ~12 ms total.*
>
> ***Average time per request** ≈ `_sum / _count` = 0.01236 s (~12 ms).*

Cardinality Explosion:

Cardinality refers to the number of unique combinations of label values (or dimensions) associated with a particular metric. Each unique combination creates a time series that must be stored and tracked over time.

Consider a metric that tracks the latency of API requests. It might have the following labels (dimensions):

- `endpoint` : The specific API endpoint being accessed (e.g., `/users` , `/orders`).
- `nodeid` : The id of the server (e.g., `node-567` , `node-343`).
- `region` : The geographical location of the server (e.g., `us-east-1` , `eu-west-1`).

For each unique combination of these labels, a new time series is created:

endpoint	nodeid	region	Latency (ms) - T1	Latency (ms) - T2
`/users`	`node-567`	`us-east-1`	120	130
`/users`	`node-343`	`us-east-1`	135	140
`/orders`	`node-567`	`us-east-1`	200	210
`/orders`	`node-343`	`us-east-1`	220	225
`/users`	`node-567`	`eu-west-1`	180	185
`/orders`	`node-567`	`eu-west-1`	190	195

If we have **1,000 endpoints** running on **100 nodes** distributed across **3 regions**, we could end up collecting **300,000 time series** at each time step.

If we were to replace `nodeid` with `user_id` , the number of unique label combinations would **skyrocket**, potentially overwhelming the monitoring system.

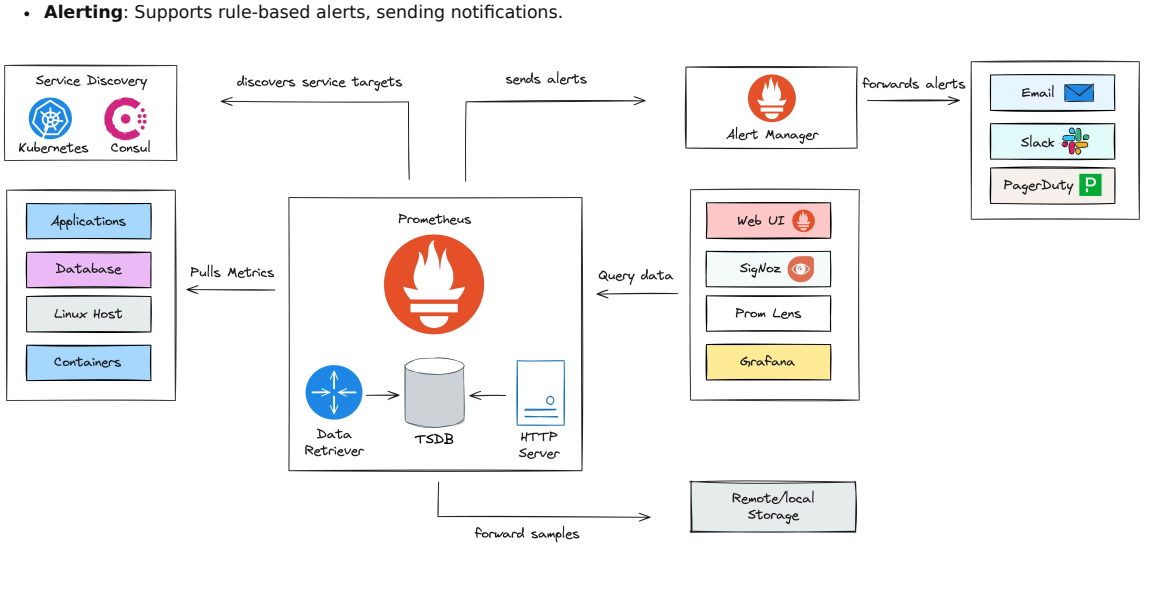
Prometheus

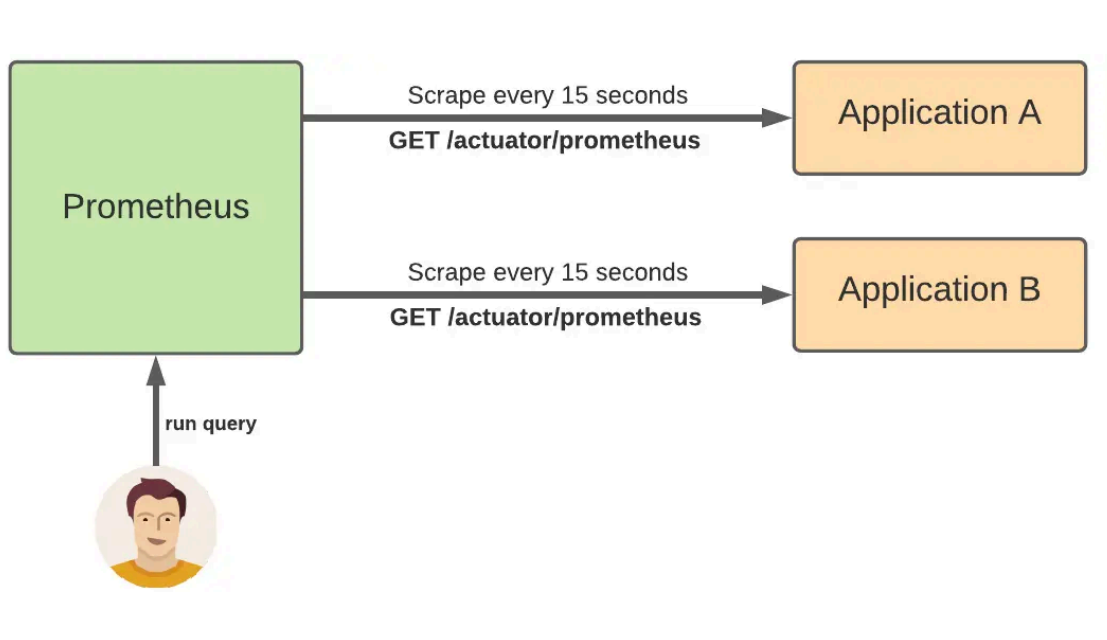
Prometheus is an open-source monitoring and alerting toolkit widely used for recording real-time metrics and generating alerts. Prometheus was developed at SoundCloud in 2012 and later became a standalone project under the umbrella of the Cloud Native Computing Foundation.

- **Service Discovery**: Automatically detects and scrapes targets using mechanisms like Kubernetes, Eureka, etc.
- **Scraping Metrics**: Collects data from targets (applications, services, systems) via HTTP at set intervals.
- **Query Language**: Provides **PromQL** for querying and analyzing time-series data.

```
# the total cumulative time spent handling HTTP server requests, measured in seconds.
http_server_request_duration_seconds_sum
rate(http_server_request_duration_seconds_sum[1m])
```

- **Alerting**: Supports rule-based alerts, sending notifications.


Service Discovery
Kubernetes
Consul
discovers service targets
sends alerts
Alert Manager
forwards alerts
Email
Slack
PagerDuty
Applications
Database
Linux Host
Containers
Pulls Metrics
Prometheus
Data Retriever
TSDB
HTTP Server
Query data
Web UI
SigNoz
Prom Lens
Grafana
forward samples
Remote/local Storage


Prometheus
Scrape every 15 seconds
GET /actuator/prometheus
Application A
Scrape every 15 seconds
GET /actuator/prometheus
Application B
run query

- **Client Libraries**: Enable applications to expose custom metrics using libraries for Go, Java, Python, and more.
- **Prometheus Server**: Core component that collects, stores, and queries metrics.
- **Data Storage**: Uses a built-in time-series database (TSDB) for efficient metric storage and retrieval.
- **Alertmanager**: Manages, deduplicates, and routes alerts to notification channels.

PromQL: Main Functions

1. Counter / Rate Functions

- `rate(v range-vector)` → Average per-second rate
- `irate(v range-vector)` → Instantaneous rate (last interval)
- `increase(v range-vector)` → Total increase over the range
- `delta(v range-vector)` → Difference between start and end value
- `resets(v range-vector)` → Number of counter resets

---

2. Gauge / Statistics Functions

- `avg_over_time(v range-vector)` → Average over the range
- `min_over_time(v range-vector)` → Minimum value over the range
- `max_over_time(v range-vector)` → Maximum value over the range
- `sum_over_time(v range-vector)` → Sum of values over the range
- `count_over_time(v range-vector)` → Number of samples
- `quantile_over_time(q, v)` → Percentile over the range

---

3. Histogram / Summary

- `histogram_quantile(q, v matrix-vector)` → Compute percentile from Histogram/Summary
- Often used with `_bucket` and `rate()`

---

4. Aggregation Functions

- `sum(v)` → Sum across series
- `avg(v)` → Average across series
- `min(v)` → Minimum across series
- `max(v)` → Maximum across series
- `count(v)` → Count of series
- `stddev(v)` → Standard deviation
- `stdvar(v)` → Variance

---

5. Transformation Functions

- `abs(v)` → Absolute value
- `ceil(v) / floor(v)` → Round up / down
- `round(v[, to_nearest])` → Round to nearest multiple
- `clamp_min(v, min) / clamp_max(v, max)` → Limit values

Metric	Function	PromQL Example	Description
HTTP Client Requests	`rate()`	`rate(http_client_request_duration_seconds_count[5m])`	Requests per second
HTTP Server Latency	`histogram_quantile()`	`histogram_quantile(0.95, sum(rate(http_server_request_duration_seconds_bucket[5m])) by (le))`	p95 request latency
JVM Memory Used	`avg_over_time()`	`avg_over_time(jvm_memory_used_bytes[10m])`	Average JVM memory used over last 10 min
JVM Threads	`max_over_time()`	`max_over_time(jvm_thread_count[5m])`	Maximum active threads
GC Duration	`rate()`	`rate(jvm_gc_duration_seconds_sum[5m])`	Average time spent in garbage collection
OTLP Exporter	`increase()`	`increase(otlp_exporter_exported_total[1h])`	Total metrics exported in 1h
Queue Size	`clamp_max()`	`clamp_max(queueSize_ratio, 1)`	Queue occupancy percentage capped at 100%
CPU Utilization	`avg_over_time()`	`avg_over_time(jvm_cpu_recent_utilization_ratio[5m])`	Average JVM CPU usage

Prometheus Problem

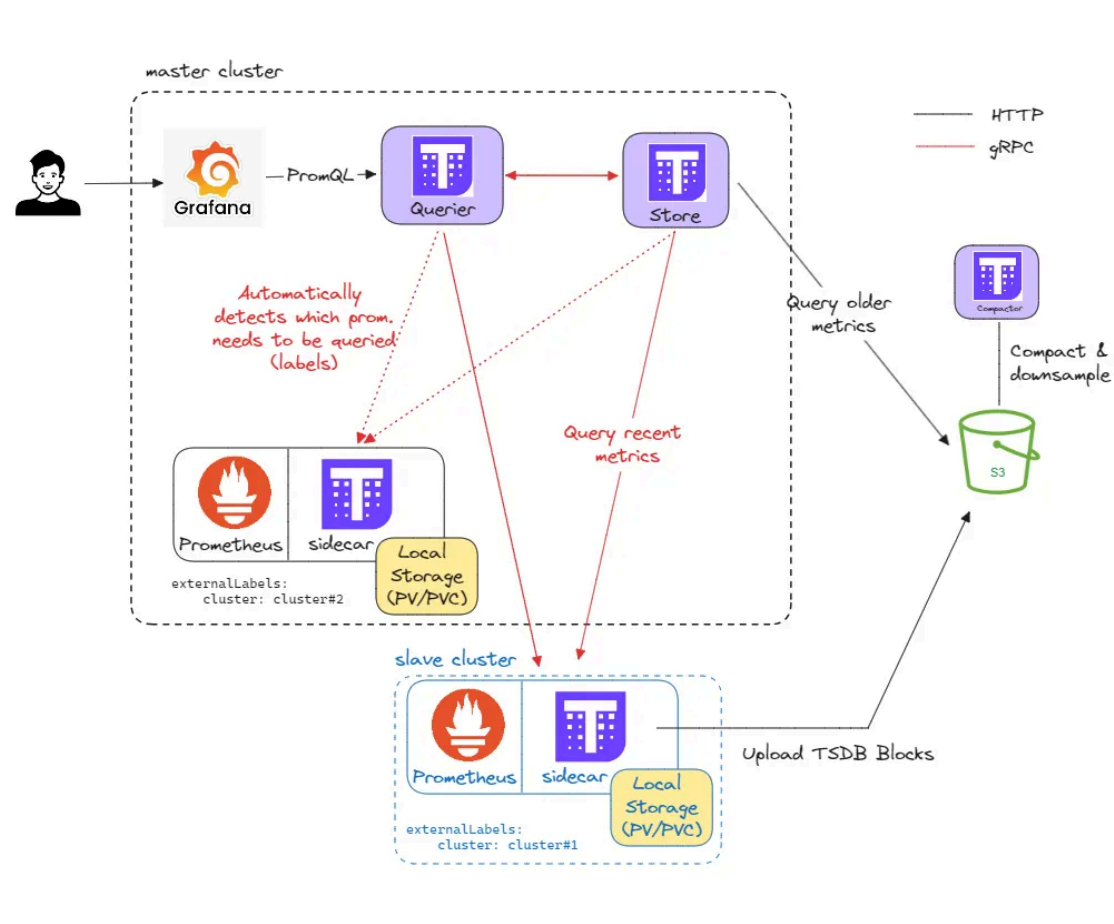
Prometheus alone is excellent for local monitoring and collecting metrics from single applications or clusters, but it has some limitations:

- **Limited vertical scalability**
  - A single Prometheus instance has limits on how much data it can write and read simultaneously.
- **Limited data retention**
  - Prometheus stores data locally in its TSDB, usually keeping metrics for only a few weeks.
- **Multi-cluster or multi-data center monitoring**
  - Prometheus is "single-node" per cluster; if you want to aggregate metrics from multiple clusters or regions, you need to run separate queries.

Thus, several projects have emerged to extend Prometheus capabilities:

- **Global View**: Aggregate metrics from multiple Prometheus instances, providing a unified and centralized query interface.
- **Long-Term Storage**: Offload Prometheus TSDB data to object storage systems (e.g., S3, GCS, Azure Blob), enabling virtually unlimited data retention.
- **High Availability**: Support replicated Prometheus instances with built-in deduplication mechanisms to prevent double-counting of metrics.

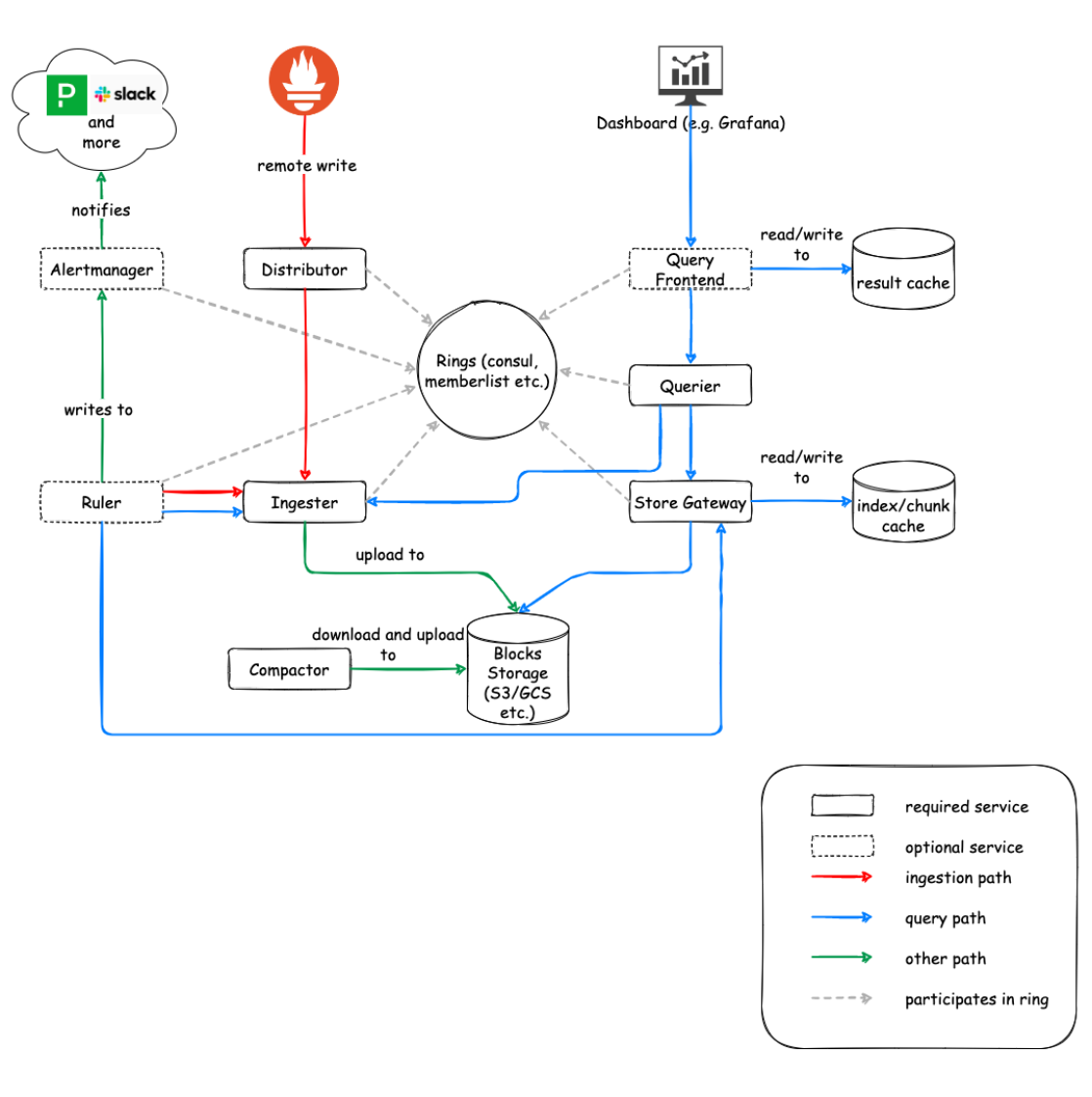
Thanos


master cluster
Grafana
PromQL
Querier
Store
HTTP
gRPC
Compactor
Compact & downsample
Query older metrics
S3
Automatically detects which prom. needs to be queried (labels)
Query recent metrics
Prometheus
sidecar
Local Storage (PV/PVC)
externalLabels:
cluster: cluster#2
slave cluster
Prometheus
sidecar
Local Storage (PV/PVC)
externalLabels:
cluster: cluster#1
Upload TSDB Blocks

- **Prometheus** feeds real-time metrics into the **Thanos Sidecar**. It also persists the data in Persistent Volume for a minimum of 6 hours.
- The **Thanos Sidecar** uploads metrics to Object Storage every 2 hours (default) and allows Thanos Query to access real-time data.
- **Thanos Store** fetches historical data from Object Storage for querying.
- **Thanos Query** aggregates data from Thanos Sidecar, Thanos Store, and potentially other query nodes to provide a unified querying interface.
- **Thanos Compactor** optimizes data in Object Storage to improve query performance.

Architecture Highlights: Uses **consistent hashing** to distribute data across store gateways and a **replication factor** to maintain multiple copies of each block. Queries are routed intelligently to only the nodes storing relevant blocks.

Cortex


slack
and more
notifies
Alertmanager
writes to
Ruler
remote write
Distributor
Ingester
Rings (consul, memberlist etc.)
upload to
download and upload to
Compactor
Blocks Storage (S3/GCS etc.)
Dashboard (e.g. Grafana)
Query Frontend
read/write to
result cache
Querier
Store Gateway
read/write to
index/chunk cache
required service
optional service
ingestion path
query path
other path
participates in ring

- **Distributor**: Receives incoming metrics, replicates them, and writes them to the correct ingesters.
- **Ingester**: Stores incoming metrics temporarily in-memory and flushes them to long-term storage.
- **Querier**: Handles read requests, fetching from ingesters and object storage.
- **Table Manager**: Manages long-term storage schemas (for DynamoDB/Cassandra).
- **Alertmanager**: Integrated or separate for alert handling.

Architecture Highlights: Uses **horizontal scaling** and **sharding via consistent hashing** to ensure load balancing. Queries intelligently target only the nodes containing the relevant time-series data.

Logs

Logs are detailed, unstructured or semi-structured textual records that describe events that occurred in the system. Logs capture the full context of operations and are the most granular observability data.

- **Rich Detail**: Logs provide detailed information about the system's state and operations, often including error messages, stack traces, and debug information.
- **Time-Stamped Events**: Each log entry is typically associated with a timestamp, allowing you to track events in chronological order.

Use Cases:

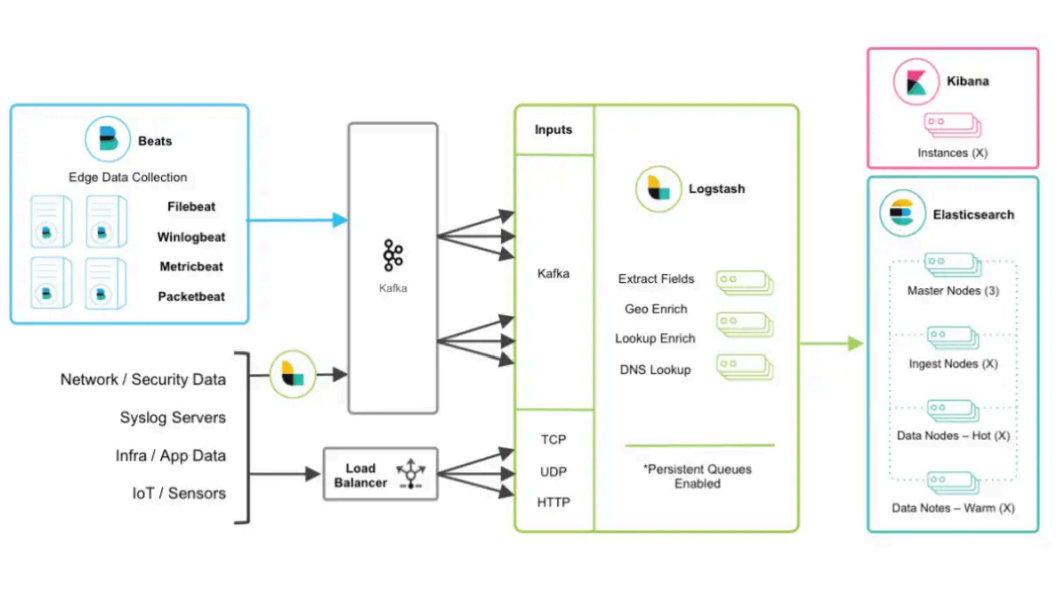
- **Debugging**: Logs are essential for troubleshooting errors and understanding the state of an application at specific points in time.
- **Auditing**: Logs can be used to track access, changes, and interactions within the system for compliance and security purposes.
- **Incident Investigation**: In the case of system failure or unexpected behavior, logs provide the information needed to reconstruct the chain of events leading to the issue.

```
[INFO] 2024-10-19 10:15:03 - Service 'OrderService' started successfully on port 8080.
[ERROR] 2024-10-19 10:16:25 - Failed to connect to the database. Error: ConnectionTimeoutException: Database not reachable at db-host:5432.
[WARN] 2024-10-19 10:18:45 - Memory usage is above the threshold (90%). Current usage: 95%.
[DEBUG] 2024-10-19 10:19:07 - Processing request for user ID 12345. Request payload: { "orderId": "9876", "product": "Laptop", "quantity": 2 }.
[TRACE] 2024-10-19 10:19:12 - Sending HTTP request to PaymentService. Endpoint: /api/payment. CorrelationId: abcd-1234-efgh-5678.
```

ELK Stack

The ELK stack is a popular set of tools used for managing and analyzing large volumes of data, particularly logs.

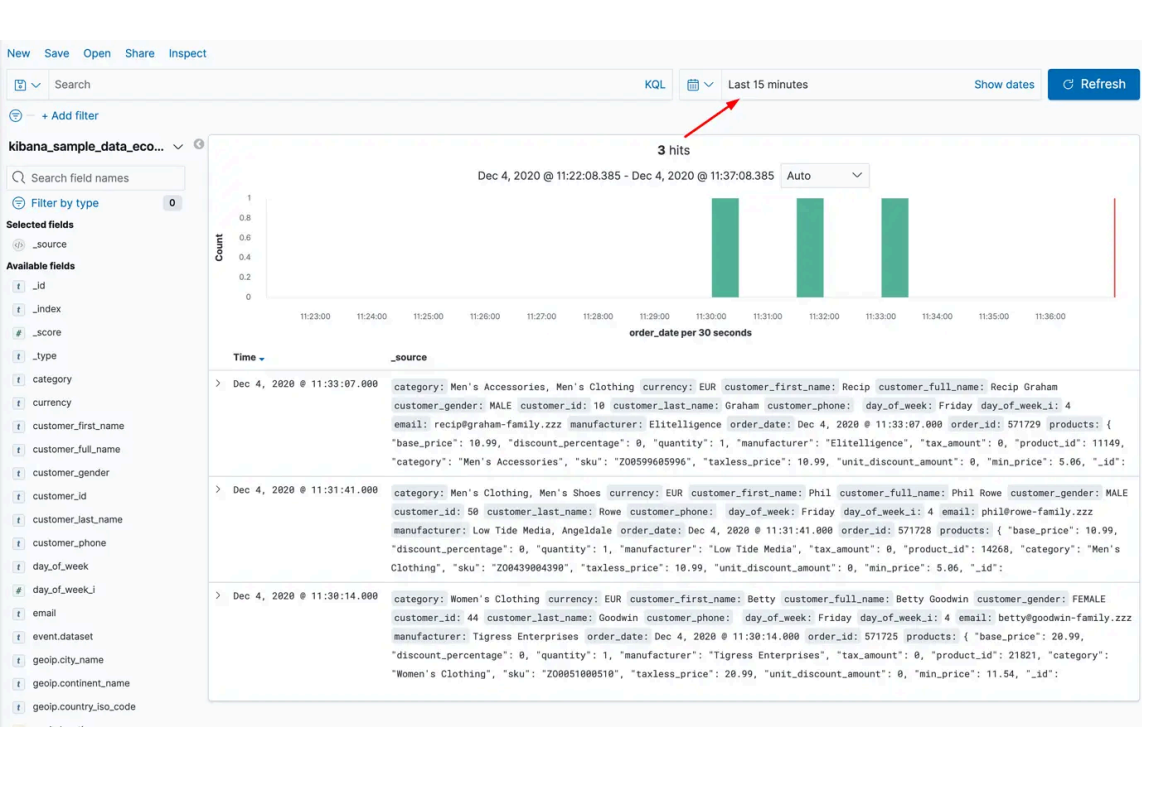
- **Centralized Logging**: The ELK stack allows organizations to centralize logs from multiple sources, making it easier to manage and analyze log data.
- **Flexible Visualization**: Kibana's visualization tools help users present data in various formats, enabling better decision-making based on insights derived from the data.
- **Scalability**: The distributed nature of Elasticsearch allows the ELK stack to scale with the growth of data, making it suitable for small to large enterprises.


Beats
Edge Data Collection
Filebeat
Winlogbeat
Metricbeat
Packetbeat
Network / Security Data
Syslog Servers
Infra / App Data
IoT / Sensors
Kafka
Load Balancer
Inputs
Kafka
TCP
UDP
HTTP
Logstash
Extract Fields
Geo Enrich
Lookup Enrich
DNS Lookup
"Persistent Queues Enabled"
Kibana
Instances (X)
Elasticsearch
Master Nodes (3)
Ingest Nodes (X)
Data Nodes – Hot (X)
Data Notes – Warm (X)

Logstash is a data pipeline that collects, transforms, and routes logs and events to destinations like Elasticsearch. It supports **multiple input sources** (e.g., files, Kafka), **filter plugins** for data processing (e.g., parsing, enrichment), and **output plugins** to forward processed data.

Elasticsearch is a distributed search and analytics engine built on Apache Lucene, enabling real-time data indexing, searching, and analysis. As the core of the ELK stack, it scales horizontally using **sharded architecture**, ensuring fault tolerance and high availability. It provides a **RESTful API** for easy integration and supports **JSON-based indexing**, allowing fast full-text searches and aggregations.

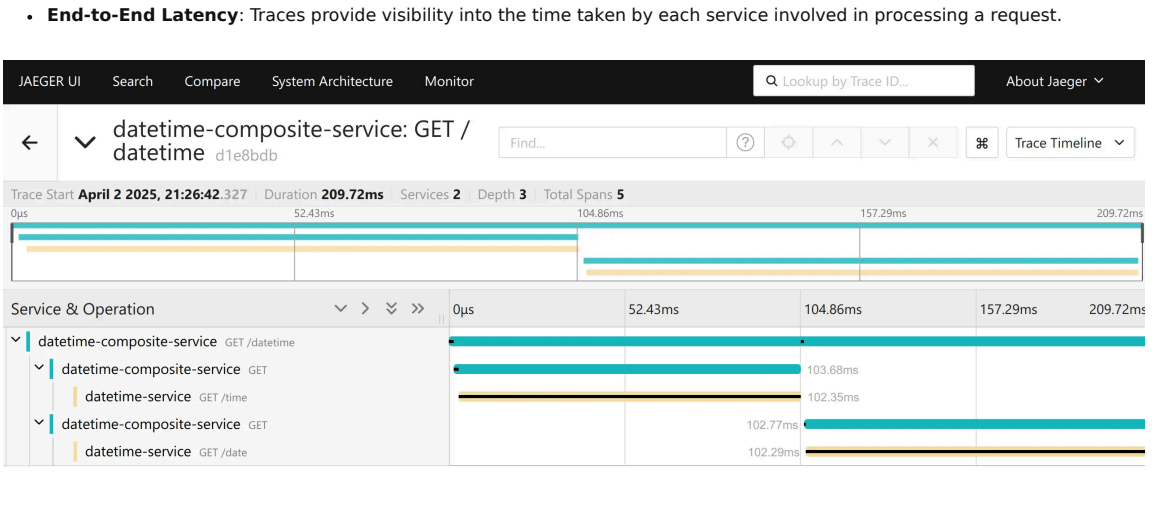
Kibana is a **visualization and analytics tool** for exploring Elasticsearch data. It allows users to create **interactive dashboards** with charts and graphs, apply **search and filtering**, and monitor logs, metrics, and traces.

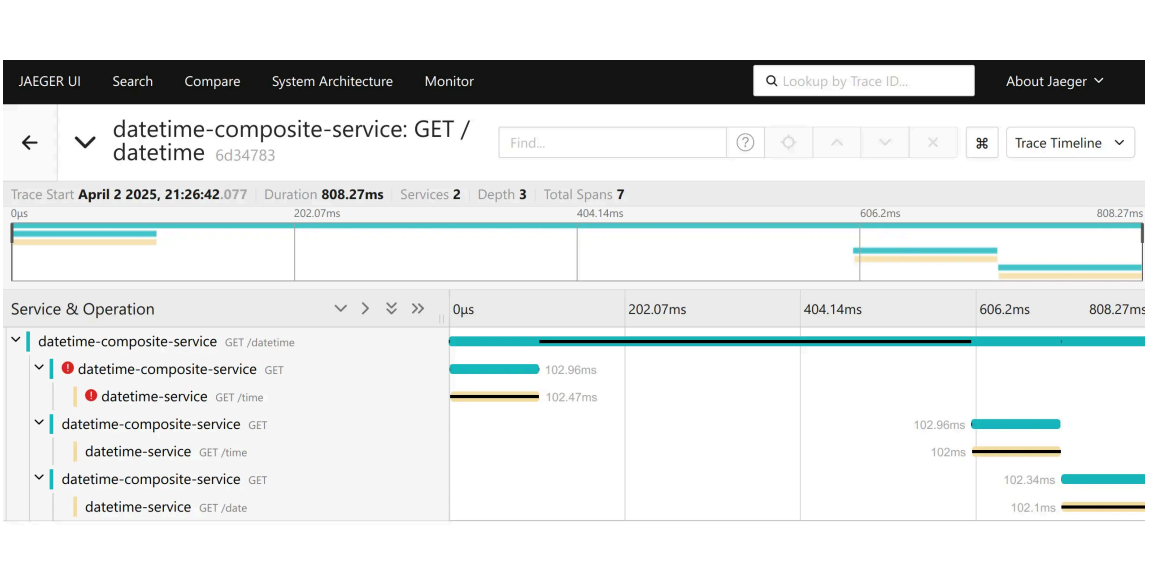

New
Save
Open
Share
Inspect
Search
KQL
Last 15 minutes
Show dates
Refresh
+ Add filter
kibana_sample_data_eco...
3 hits
Dec 4, 2020 @ 11:22:08.385 - Dec 4, 2020 @ 11:37:08.385
Auto
Count
order_date per 30 seconds
Time
_source
Search field names
Filter by type
Selected fields
Available fields

Traces

Traces track the path of a request as it moves through various services in a distributed system. They help visualize how requests propagate across different components (a sort of distributed stack trace).

- **Span and Trace IDs**: Traces are composed of spans, which represent a single operation within a service. Each span contains a unique ID, and all spans related to a single request share the same trace ID.
- **End-to-End Latency**: Traces provide visibility into the time taken by each service involved in processing a request.


JAEGER UI
Search
Compare
System Architecture
Monitor
Lookup by Trace ID...
About Jaeger
datetime-composite-service: GET / datetime d1e8bdb
Trace Timeline
Trace Start April 2 2025, 21:26:42.327
Duration 209.72ms
Services 2
Depth 3
Total Spans 5
Service & Operation
0μs
52.43ms
104.86ms
157.29ms
209.72ms
datetime-composite-service GET /datetime
datetime-composite-service GET
datetime-service GET /time
datetime-composite-service GET
datetime-service GET /date

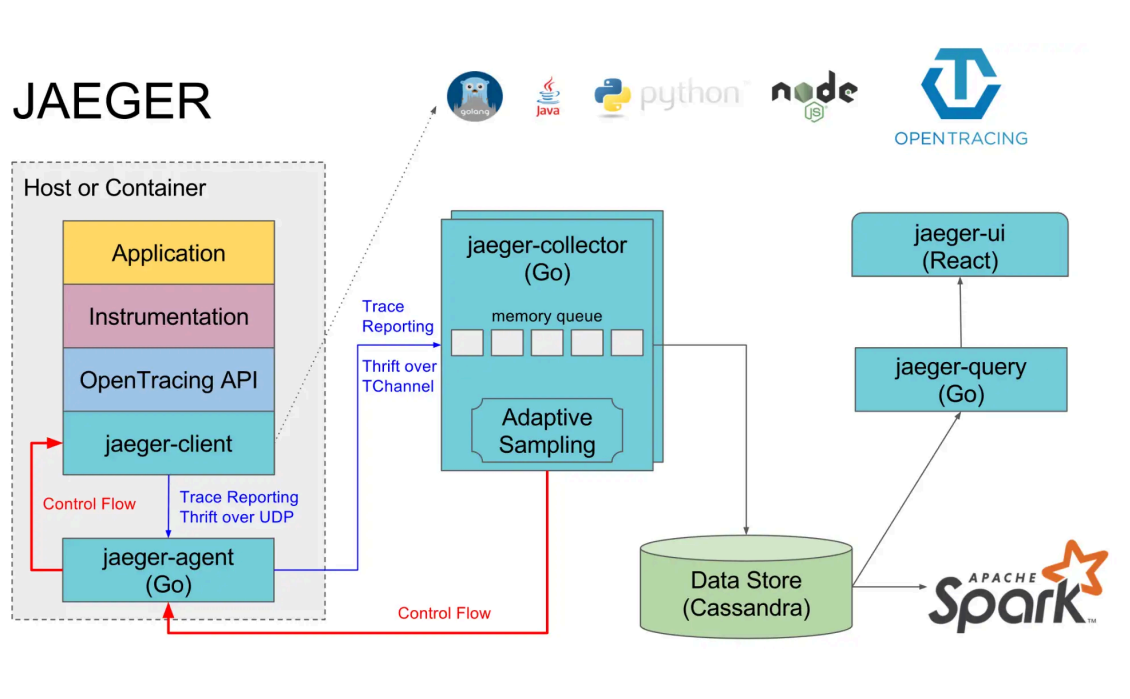

JAEGER UI
Search
Compare
System Architecture
Monitor
Lookup by Trace ID...
About Jaeger
datetime-composite-service: GET / datetime 6d34783
Trace Timeline
Trace Start April 2 2025, 21:26:45.077
Duration 808.27ms
Services 2
Depth 3
Total Spans 7
Service & Operation
0μs
202.07ms
404.14ms
606.2ms
808.27ms
datetime-composite-service GET /datetime
datetime-composite-service GET
datetime-service GET /time
datetime-composite-service GET
datetime-service GET /time
datetime-composite-service GET
datetime-service GET /date

Use Cases:

- **Distributed Context Propagation**: Tracks requests as they propagate across multiple services, providing visibility into their interactions.
- **Latency Analysis**: Enables analysis of request timing to identify performance bottlenecks and optimize service interactions.
- **Performance Optimization**: Helps detect bottlenecks by showing how long each service takes to process a request.
- **Root Cause Analysis**: Makes it possible to pinpoint which service is responsible for slowdowns or errors.
- **Dependency Visualization**: Provides a clear view of how services interact, making complex dependencies easier to understand in a microservice architecture.

Traces: Jaeger/Zipkin

Jaeger and Zipkin are an open-source end-to-end distributed tracing system designed for monitoring and troubleshooting the performance of microservices-based architectures.


JAEGER
Java
python
node
OPENTRACING
Host or Container
Application
Instrumentation
OpenTracing API
jaeger-client
Control Flow
Trace Reporting Thrift over UDP
jaeger-agent (Go)
jaeger-collector (Go)
memory queue
Trace Reporting
Thrift over TChannel
Adaptive Sampling
Control Flow
Data Store (Cassandra)
jaeger-ui (React)
jaeger-query (Go)
APACHE Spark

Combining Metrics, Logs, and Traces

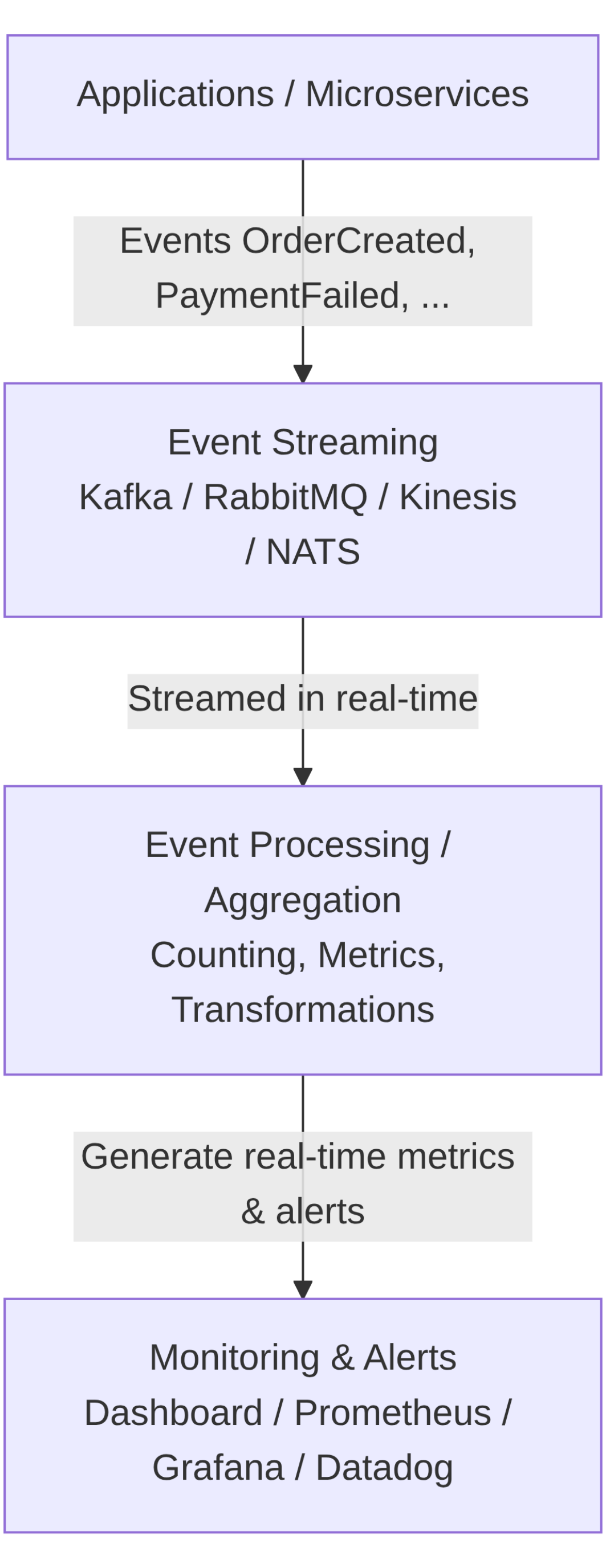
Although each pillar serves a different purpose, they complement one another to provide full visibility into system health:

- **Metrics** indicate **what** is happening, such as high CPU usage or slow response times.
- **Logs** provide the **why** by capturing detailed event data preceding or occurring during an issue.
- **Traces** illustrate the **how** by mapping request flows across system components, enabling the identification of performance bottlenecks or failures.

By integrating these observability pillars with specialized tools—such as Prometheus for metrics, the ELK stack for logs, and Jaeger for tracing—teams can gain comprehensive visibility into system behavior. This integration is crucial for effective monitoring, debugging, and optimization of modern software systems.

Event-Based Observability

In addition to metrics, logs, and traces, modern observability leverages **real-time event streams** to gain immediate insights.


Applications / Microservices
Events OrderCreated, PaymentFailed, ...
Event Streaming
Kafka / RabbitMQ / Kinesis / NATS
Streamed in real-time
Event Processing / Aggregation
Counting, Metrics, Transformations
Generate real-time metrics & alerts
Monitoring & Alerts
Dashboard / Prometheus / Grafana / Datadog

- **Event Stream Sources**
  - Messaging systems such as **Apache Kafka**, **RabbitMQ**, **AWS Kinesis**, or **NATS** capture events from applications or microservices.
  - Events may represent completed operations, errors, state changes, or user interactions.
- **Real-Time Metrics and Alerts**
  - Events can be aggregated into metrics or triggers in near real-time without waiting for polling intervals.
  - Example: Counting `OrderCreated` events in Kafka to monitor throughput and detect sudden drops or spikes.

Instrumentation

Instrumentation in microservices refers to the process of collecting metrics, logs, and traces to monitor, diagnose, and improve performance. Instrumenting microservices involves costs in terms of resource consumption, performance overhead, and engineering effort.

The B2I (Business Logic to Instrumentation) ratio

To calculate the B2I ratio, determine the number of lines of code (LOC) before adding an instrumentation (adding code for emitting signals for a signal type), and then determine the LOC after the instrumentation. The B2I ratio is then:

B2I = LOC_AFTER_INSTR / LOC_BEFORE_INSTR

In an ideal world, the B2I ratio would be 1, representing zero instrumentation costs in the code. However, in reality, the more LOC you dedicate to instrumentation, the higher the B2I ratio is. For example, if your code has 3800 LOC and you added 400 LOC for instrumentation (say, to emit logs and metrics), then you'd end up with a B2I ratio of 1.105, from (3800 + 400) / 3800.

Manual instrumentation

Metrics

Micrometer is a popular metrics collection library in the Spring ecosystem, often used with Prometheus or other monitoring tools.

```
import io.micrometer.core.instrument.MeterRegistry;
import io.micrometer.core.instrument.Counter;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;

@RestController
public class ExampleController {

    private final Counter requestCounter;

    public ExampleController(MeterRegistry meterRegistry) {
        this.requestCounter = meterRegistry.counter("http.requests.count");
    }

    @GetMapping("/hello")
    public String hello() {
        requestCounter.increment();  // Increment the metric counter
        return "Hello, World!";
    }
}
```

```
from prometheus_client import Counter, start_http_server
from flask import Flask

# Create a Counter metric
REQUEST_COUNTER = Counter('http_requests_total', 'Total HTTP requests')

app = Flask(__name__)

@app.route("/hello")
def hello():
    REQUEST_COUNTER.inc()  # Increment the metric counter
    return "Hello, World!"

if __name__ == "__main__":
    # Start Prometheus metrics server on port 8000
    start_http_server(8000)
    app.run(port=5000)
```

Logs

SLF4J (Simple Logging Facade for Java) is a popular logging abstraction in the Java ecosystem, commonly used with logging frameworks like **Logback** or **Log4j** to provide a flexible and consistent logging API.

```
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;

@RestController
public class LoggingController {

    private static final Logger logger = LoggerFactory.getLogger(LoggingController.class);

    @GetMapping("/data")
    public String fetchData() {
        logger.info("Fetching data...");
        // Simulate data retrieval
        return "Data retrieved!";
    }
}
```

```
import logging
from flask import Flask

# Configure logging
logging.basicConfig(
    level=logging.INFO,
    format='%(asctime)s [%(levelname)s] %(name)s - %(message)s'
)
logger = logging.getLogger("LoggingController")

app = Flask(__name__)

@app.route("/data")
def fetch_data():
    logger.info("Fetching data...")
    # Simulate data retrieval
    return "Data retrieved!"

if __name__ == "__main__":
    app.run(port=5000)
```

The logging system has to be configured to send logs remotely.

```
<?xml version="1.0" encoding="UTF-8"?>
<configuration>

    <appender name="STDOUT" class="ch.qos.logback.core.ConsoleAppender">
        <encoder>
            <pattern>%d{dd-MM-yyyy HH:mm:ss.SSS} [%thread] %-5level %logger{36}.%M - %msg%n</pattern>
        </encoder>
    </appender>

    <appender name="FILE" class="ch.qos.logback.core.FileAppender">
        <file>logs/application.log</file>
        <encoder>
            <Pattern>%d{dd-MM-yyyy HH:mm:ss.SSS} [%thread] %-5level %logger{36}.%M - %msg%n</Pattern>
        </encoder>
    </appender>

    <appender name="LOKI" class="com.github.loki4j.logback.Loki4jAppender">
        <http>
            <url>${LOKI_ENDPOINT}</url>
        </http>
        <format>
            <label>
                <pattern>app=datetime-service,host=${HOSTNAME}</pattern>
            </label>
            <message>
                <pattern>%-5level [%.5(${HOSTNAME})] %.10thread %logger{20} | %msg %ex</pattern>
            </message>
        </format>
    </appender>

    <root level="INFO">
        <appender-ref ref="STDOUT"/>
        <appender-ref ref="FILE"/>
        <appender-ref ref="LOKI"/>
    </root>
</configuration>
```

Traces

OpenTelemetry is a widely adopted observability framework that provides **tracing, metrics, and logging** capabilities.

```
import io.opentelemetry.api.trace.Span;
import io.opentelemetry.api.trace.Tracer;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;

@RestController
public class TracingController {

    private final Tracer tracer;

    public TracingController(Tracer tracer) {
        this.tracer = tracer;
    }

    @GetMapping("/process")
    public String processRequest() {
        Span span = tracer.spanBuilder("processRequest").startSpan();  // Start tracing span
        try {
            // Simulate some business logic
            Thread.sleep(500);
            return "Processed";
        } catch (InterruptedException e) {
            span.recordException(e);
            return "Failed";
        } finally {
            span.end();  // End the span
        }
    }
}
```

```
# Configure OpenTelemetry tracer
resource = Resource(attributes={"service.name": "tracing-service"})
trace.set_tracer_provider(TracerProvider(resource=resource))
tracer = trace.get_tracer(__name__)

# Export spans to console (for demo purposes)
span_processor = BatchSpanProcessor(ConsoleSpanExporter())
trace.get_tracer_provider().add_span_processor(span_processor)

app = Flask(__name__)

@app.route("/process")
def process_request():
    with tracer.start_as_current_span("processRequest") as span:
        try:
            # Simulate some business logic
            time.sleep(0.5)
            return "Processed"
        except Exception as e:
            span.record_exception(e)
            return "Failed"

if __name__ == "__main__":
    app.run(port=5000)
```

Zero Code Instrumentation

Definition:

- Technique where **no changes to application code** are needed to enable monitoring, observability, or performance tracking.
- Achieved using **automatic instrumentation** via agents or frameworks.
- **Supported Frameworks**
  - Java: Spring Boot, gRPC, JDBC, HTTP clients
  - Python: Flask, Django, requests, SQLAlchemy, Celery
  - Node.js: Express, HTTP, MySQL/Postgres, Redis

What is a Java Agent?

- Special Java program that can **intercept and modify bytecode at runtime**
- Uses the **Java Instrumentation API**
- Injects **custom behavior** into classes **before or during loading** by the JVM

Command to attach an agent:

```
java -javaagent:/path/to/agent.jar -jar your-application.jar
```

How It Works

1. **Agent Startup**

- JVM loads the agent before the main application
- `premain()` method is invoked

2. **Class Transformation**

- JVM provides the **bytecode** of each class as it is loaded
- Agent can **modify the bytecode** (inject logging, metrics, tracing)

Intercepting Methods with Byte Buddy

Original Method

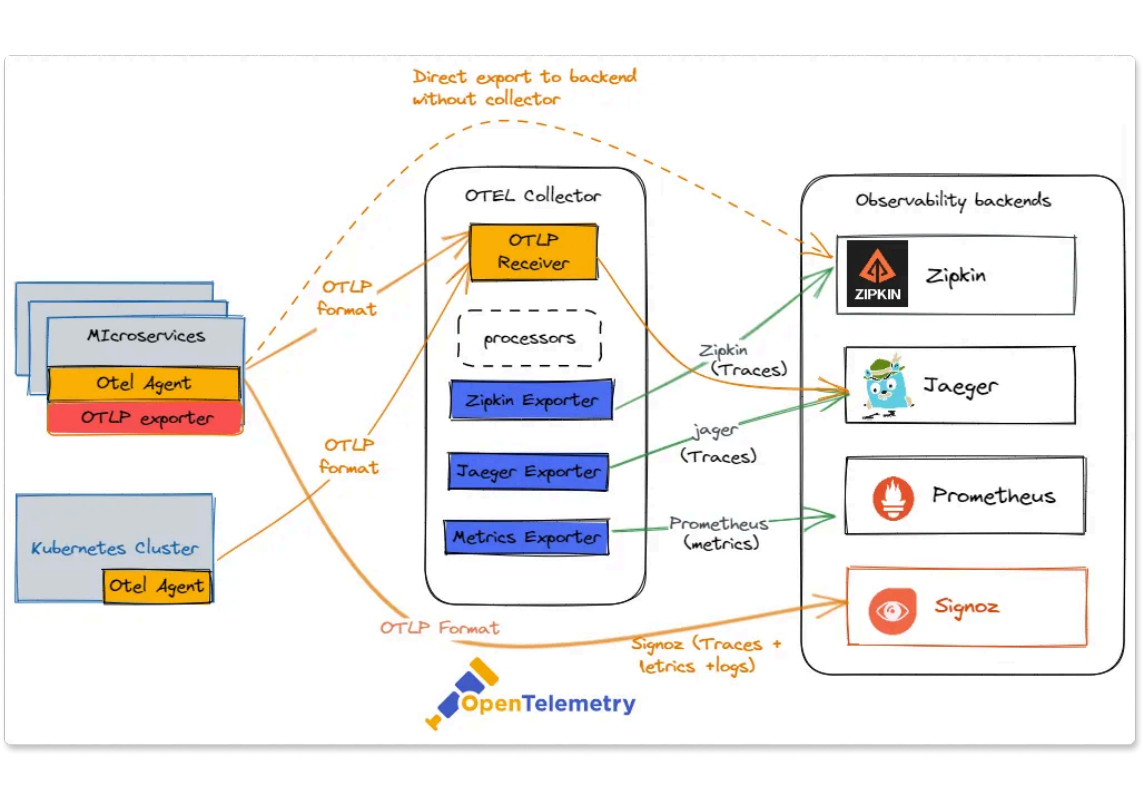
```
public void doWork() {
    businessLogic();
}
```

Instrumented Method via Agent

```
public void doWork() {
    long start = System.currentTimeMillis();
    try {
        businessLogic();
    } finally {
        long end = System.currentTimeMillis();
        log(end - start);
    }
}
```

```
public class LoggerInterceptor {
    @RuntimeType
    public static Object intercept(@SuperCall Callable<?> zuper) throws Exception {
        long start = System.currentTimeMillis();
        try {
            return zuper.call(); // original method executes
        } finally {
            long end = System.currentTimeMillis();
            System.out.println("Execution time: " + (end - start) + "ms");
        }
    }
}
```

OpenTelemetry Collector (Universal Telemetry Agent)


Direct export to backend without collector
MIcroservices
Otel Agent
OTLP exporter
Kubernetes Cluster
Otel Agent
OTLP format
OTLP format
OTEL Collector
OTLP Receiver
processors
Zipkin Exporter
Jaeger Exporter
Metrics Exporter
Zipkin (Traces)
jager (Traces)
Prometheus (metrics)
OTLP Format
Signoz (Traces + letrics +logs)
Observability backends
Zipkin
ZIPKIN
Jaeger
Prometheus
Signoz
OpenTelemetry

The **OpenTelemetry Collector** is an open-source, vendor-neutral tool for collecting, processing, and exporting **logs, metrics, and traces**. It provides a **unified data collection** solution, replacing multiple specialized agents, and can be deployed either as a local agent or a central service. Its **extensible pipeline** includes:

- **Receivers**: ingest data from various sources.
- **Processors**: enrich, filter, or transform data.
- **Exporters**: send data to backends like Prometheus, Elasticsearch, Grafana, or third-party platforms.

It offers **scalability, flexibility, and portability** across different infrastructures.

Observability as a Platform

Modern observability favors **unified platforms** over isolated tools.

- **Grafana Cloud** – Integrates metrics, logs, and tracing with Tempo, Loki, Mimir.
- **SigNoz** – Single datastore (ClickHouse) for metrics, logs, traces; simpler to operate than separate stacks.
- **OpenObserve** – Lightweight Rust-based architecture, optimized for cost, performance, and cloud-native environments.
- **Benefits**
  - Simplifies operational complexity.
  - Centralizes data collection.
  - Reduces inconsistency across multiple tools.

Observability Costs

- **Storage**: High-volume or high-cardinality data is costly; retention policies and compression help manage expenses.
- **Compute**: Processing tasks like queries, aggregations, and anomaly detection increase compute costs at scale.
- **Networking**: Cross-region transfers, streaming analytics, and API calls can add significant network costs.
- **Licensing & Tooling**: Proprietary tools incur license fees; open-source tools require infrastructure and maintenance resources.

Best Practices

- **Control Cardinality** - Avoid highly granular labels (e.g., `user_id`) in metrics to prevent time series explosion.
- **Data Privacy** - Never log sensitive data (passwords, PII).
- **Scraping Frequency and Overhead** - Balance frequency of metric collection and log verbosity with system resource consumption.
- **Retention Policies** - Define retention and compression policies to control storage costs while retaining useful data for analysis.

Resources

- Cloud Observability in Action, Hausenblas